

ITM SLS BARODA UNIVERSITY, VADODARA
B.TECH. SEMESTER-3 (BRANCH – ALL; SCHOOL – SOCSET)
DEPARTMENT OF APPLIED MATHEMATICS (S.O.S.)
COMPUTER ORIENTED NUMERICAL METHODS WITH PYTHON (S2320C1)

TUTORIAL – SOLUTION OF SYSTEM OF LINEAR EQUATIONS

Part-A: Multiple Choice Questions

- Q-1. Gauss-Jacobi method is applicable if the system matrix is
(a) Sparse (b) Diagonally dominant (c) Singular (d) All mentioned
- Q-2. The Gauss-Seidel method converges faster than Gauss-Jacobi because
(a) It updates all values at once (b) It does not require initial guess
(c) It uses latest available values (d) All mentioned
- Q-3. Gauss-Seidel iterative method is used to solve
(a) Differential equation (b) System of non-linear equation
(c) System of linear equation (d) Partial differential equations
- Q-4. Gauss-Seidel method gives result faster when the pivotal element are
(a) Smaller than other coefficient (b) Larger than other coefficient
(c) Equal to other coefficient (d) None of the mentioned
- Q-5. Gauss Seidel method is also termed as a method of
(a) Iterations (b) False positions (c) Successive displacement (d) Eliminations

Part-B: Theory Questions

- Q-1. List the Direct Methods and the Iterative Methods of solving system of simultaneous linear equations.
- Q-2. State at least two differences between Direct Methods and Iterative Methods.
- Q-3. State the sufficient condition for convergence of Gauss-Seidel and Gauss-Jacobi iteration methods.
- Q-4. State at least two points to compare Gauss-Seidel and Jacobi iteration methods.
- Q-5. Explain ill-conditioned equations. Check whether the following system is ill-conditioned or not: $1.01x + 2y = 2.01$, $x + 2y = 2$

Part-C: Numerical Questions

- Q-1. Use Jacobi Iteration to solve the following systems:
(a) $5x - y + z = 10$, $2x + 4y = 12$, $x + y + 5z = -1$ Correct to three decimal places.
Ans: $x = 2.556$, $y = 1.722$, $z = -1.055$

(b) $10x + y - z = 11.19$, $x + 10y + z = 28.08$, $-x + y + 10z = 35.61$ Correct to two decimals.

Ans: $x = 1.23$, $y = 2.34$, $z = 3.45$

(c) $x + y + 54z = 110$, $27x + 6y - z = 12$, $6x + 15y + 2z = 72$ Correct to four decimal places.

Ans: $x = 2.4255$, $y = 3.5730$, $z = 1.9260$

(d) $x - 2y + 10z = 30.6$, $2x + 5y - z = 10.5$, $3x + y + z = 9.3$ Correct to four decimals.

Ans: $x = 1.2109$, $y = 2.2939$, $z = 3.3977$

(e) $2x - 3y + 10z = 3$, $-x + 4y + 2z = 20$, $5x + 2y + z = -12$ Correct to four decimals.

Ans: $x = -3.8579$, $y = 2.9848$, $z = 1.88656$

Q-2. Use Seidel Iteration to solve the following systems:

(a) $28x + 4y - z = 32$, $x + 3y + 10z = 24$, $2x + 17y + 4z = 35$ Correct to three decimals.

Ans: $x = 0.998$, $y = 1.723$, $z = 2.024$

(b) $7x_1 + 52x_2 + 13x_3 = 104$, $83x_1 + 11x_2 - 4x_3 = 95$, $3x_1 + 8x_2 + 29x_3 = 71$ Correct to three decimal places.

Ans: $x_1 = 1.058$, $x_2 = 1.367$, $x_3 = 1.962$

(c) $8x - 3y + 2z = 20$, $6x + 3y + 12z = 35$, $4x + 11y - z = 33$ Correct to four decimals.

Ans: $x = 3.0167$, $y = 1.9858$, $z = 0.9118$

(d) $3x + 4y + 15z = 54.8$, $x + 12y + 3z = 39.66$, $10x + y - 2z = 7.74$ Correct to four decimal

Ans: $x = 1.075$, $y = 2.524$, $z = 2.765$

(e) $30x - 2y + 3z = 75$, $x + 17y - 2z = 48$, $x + y + 9z = 15$ Correct to two decimal places.

Ans: $x = 2.58$, $y = 2.80$, $z = 1.07$